Supplementary Materials: Tables S1–S6

Comparative Genomics of Longevity, Encephalization, and Cancer Resistance in Psittaciformes
(with 71-Species Extension and Full Red-Team Statistical Audit)

Angelo — UNICEPLAC, Brasília, DF, Brazil
Repository: `github.com/redkk123/Parrots` (`claude/new-session-lwh7xh`)

November 10, 2026

Contents

Table S1: Validated 71-Species Cohort **2**

Table S2: H8 58 vs. 71 Species & Telomere Sensitivity **4**

Table S3: Modular Null & aBSREL Base Rate **5**

Table S4: Red-Team Sensitivity of Phase 3 Coupling **6**

Table S5: Tier A vs. Tier B Candidate Genes **8**

Table S6: Statistical Inventory & 3D Structural Summary **9**

Table S1. Validated 71-Species Genomic Cohort from 126 Audited Assemblies

Complete list of all 71 included Psittaciformes species passing the pre-registered Phase 1 audit ($\text{BUSCO}_{\text{complete}} \geq 70\%$ on `compleasm aves_odb10`, nuclear Mash identity confirmed, deduplicated to one best assembly per species). Cohort indicates the frozen 58-species discovery panel vs. the 13 newly finalized assemblies (`holdout_13`).

Species	Cohort	Clade	Accession	Source	Co
Agapornis canus	Discovery (58)	Psittaculidae	GCA_036873685.1	Iridian	
Agapornis lilianae	Discovery (58)	Psittaculidae	GCA_035222285.1	Iridian	
Agapornis personatus	Discovery (58)	Psittaculidae	GCA_034783155.1	Iridian	
Cyanoramphus novaezelandiae	Discovery (58)	Psittaculidae	GCA_025629965.1	Iridian	
Lathamus discolor	Discovery (58)	Psittaculidae	GCA_037157625.1	Vertebrate Genomes Pro	
Loriculus galgulus	Discovery (58)	Psittaculidae	GCA_046563285.1	Harvrad University	
Lorius chlorocercus	Discovery (58)	Psittaculidae	GCA_025629885.1	Iridian	
Lorius garrulus	Discovery (58)	Psittaculidae	GCA_010014925.2	Iridian	
Lorius lory	Discovery (58)	Psittaculidae	GCA_025629905.1	Iridian	
Melopsittacus undulatus	Discovery (58)	Psittaculidae	GCA_012275295.1	G10K	
Neopsephotus bourkii	Discovery (58)	Psittaculidae	GCA_033783945.1	CSIRO Applied Genomics	
Platycercus adscitus	Discovery (58)	Psittaculidae	GCA_025629925.1	Iridian	
Psephotus haematonotus	Discovery (58)	Psittaculidae	GCA_026283765.1	Iridian	
Psittacula derbiana	Discovery (58)	Psittaculidae	GCA_029814625.1	Iridian	
Trichoglossus euteles	Discovery (58)	Psittaculidae	GCA_025447915.1	Iridian	
Amazona aestiva	Discovery (58)	Other	GCA_017639355.1	University of Vienna	
Amazona finschi	Discovery (58)	Other	GCA_031763935.1	Iridian	
Amazona vinacea	Discovery (58)	Other	GCA_026283615.1	Iridian	
Ara ambiguus	Discovery (58)	Other	GCA_026284625.1	Iridian	
Ara ararauna	Discovery (58)	Other	GCA_028858755.1	Vertebrate Genomes Pro	
Ara rubrogenys	Discovery (58)	Other	GCA_917880015.2	IBE (Institut de Biolo	
Aratinga auricapillus	Discovery (58)	Other	GCA_026284685.1	Iridian	
Aratinga solstitialis	Discovery (58)	Other	GCA_902168055.1	INSTITUT DE BIOLOGIA E	
Bolborhynchus lineola	Discovery (58)	Other	GCA_025448015.1	Iridian	
Brotogeris tirica	Discovery (58)	Other	GCA_034782575.1	Iridian	
Cacatua alba	Discovery (58)	Other	GCA_024361745.1	Iridian	
Cacatua ducorpsii	Discovery (58)	Other	GCA_025448155.1	Iridian	
Cacatua galerita	Discovery (58)	Other	GCA_035583095.1	Iridian	
Cacatua moluccensis	Discovery (58)	Other	GCA_025448035.1	Iridian	
Cacatua sanguinea	Discovery (58)	Other	GCA_030265285.1	Iridian	
Cacatua sulphurea	Discovery (58)	Other	GCA_031762475.1	Iridian	
Cyanoliseus patagonus	Discovery (58)	Other	GCA_055772155.1	Bird 10,000 Genomes (B	
Deroptyus accipitrinus	Discovery (58)	Other	GCA_026284505.1	Iridian	
Diopsittaca nobilis	Discovery (58)	Other	GCA_024337085.1	Iridian	
Eolophus roseicapilla	Discovery (58)	Other	GCA_013397615.1	B10K Consortium	
Eupsittula nana	Discovery (58)	Other	GCA_026284785.1	Iridian	
Eupsittula pertinax	Discovery (58)	Other	GCA_011317185.2	Iridian	
Guaruba guarouba	Discovery (58)	Other	GCA_045345405.1	Genomics of Brazilian	
Myiopsitta monachus	Discovery (58)	Other	GCA_017639245.1	University of Vienna	
Nestor notabilis	Discovery (58)	Other	GCA_000696875.1	BGI	
Nymphicus hollandicus	Discovery (58)	Other	GCA_009823435.2	Iridian	
Pionites melanocephalus	Discovery (58)	Other	GCA_029814565.1	Iridian	
Poicephalus gulielmi	Discovery (58)	Other	GCA_982290535.1	B10K project	
Poicephalus rueppellii	Discovery (58)	Other	GCA_031762375.1	Iridian	
Poicephalus senegalus	Discovery (58)	Other	GCA_025448095.1	Iridian	
Primolius auricollis	Discovery (58)	Other	GCA_026284745.2	Iridian	
Primolius couloni	Discovery (58)	Other	GCA_024331605.1	Iridian	
Psilopsiagon aurifrons	Discovery (58)	Other	GCA_025447955.1	Iridian	
Psilopsiagon aymara	Discovery (58)	Other	GCA_024733675.1	Iridian	
Psittacara acuticaudatus	Discovery (58)	Other	GCA_026284765.1	Iridian	
Psittacara holochlorus	Discovery (58)	Other	GCA_024320895.1	Iridian	
Psittacara leucophthalmus	Discovery (58)	Other	GCA_024331625.1	Iridian	
Psittacara wagleri	Discovery (58)	Other	GCA_024361765.2	Iridian	

Species	Cohort	Clade	Accession	Source	Co
Psittacus erithacus	Discovery (58)	Other	GCA_009867235.2	Iridian	
Psittacus timneh	Discovery (58)	Other	GCA_009867315.2	Iridian	
Pyrrhura griseipectus	Discovery (58)	Other	GCA_010014965.2	Iridian	
Pyrrhura leucotis	Discovery (58)	Other	GCA_038406155.1	Iridian	
Rhynchopsitta pachyrhyncha	Discovery (58)	Other	GCA_026284805.1	Iridian	
Agapornis fischeri	Holdout (13)	Psittaculidae	GCA_051622115.1	Southwest University	
Agapornis nigrigenis	Holdout (13)	Psittaculidae	GCA_988293565.1	WELLCOME SANGER INSTIT	
Agapornis roseicollis	Holdout (13)	Psittaculidae	GCA_051622155.1	Southwest University	
Neophema chrysogaster	Holdout (13)	Psittaculidae	GCA_047054915.1	The University of Sydn	
Psittacula krameri	Holdout (13)	Psittaculidae	GCA_988280915.1	WELLCOME SANGER INSTIT	
Amazona guildingii	Holdout (13)	Other	GCA_013399615.1	B10K Consortium	
Amazona ochrocephala	Holdout (13)	Other	GCA_039720435.1	Vertebrate Genomes Pro	
Amazona vittata	Holdout (13)	Other	GCA_980694505.1	institute for systems	
Forpus passerinus	Holdout (13)	Other	GCA_981469625.1	B10K project	
Probosciger aterrimus	Holdout (13)	Other	GCA_013397665.1	B10K Consortium	
Psittacara erythrogenys	Holdout (13)	Other	GCA_982290595.1	B10K project	
Psittacara finschi	Holdout (13)	Other	GCA_982290525.1	B10K project	
Psittrichas fulgidus	Holdout (13)	Other	GCA_055772115.1	Bird 10,000 Genomes (B	

Table S2. Phase 2 H8 Confirmatory Selection Tests (58 vs. 71 Species) and Telomere Sensitivity Audit

Comparison of key H8 candidate genes across the frozen 58-species cohort and the extended 71-species cohort (`burgio2019` topology), alongside Red-Team alignment sensitivity (MAFFT vs. L-INS-i vs. stringent window/gap cleaning vs. `aBSREL` `-multiple-hits Double+Triple`) and *Strigops habroptila* (kakapo) dual-reference orthology concordance.

Gene	**Mod.**	$N_{58/71}$
ACD	M7	6/11
POT1	M7	8/26
RTEL1	M7	23/35
TERF1	M7	9/25
TERF2	M7	18/29
TERT	M7	15/34
DKC1	M7	49/60
MRE11	M4	49/63
SOD1	M6	54/67
NQO1	M5	41/56
BBC3	M3	36/52
CCNE1	M2	49/62
MYC	M2	5/6
APEX1	M4	3/8
APAF1	M3	17/33

Table S3. Phase 2 Modular Null Benchmark (298 Control Genes) and Empirical aBSREL Base Rate

Top panel: H8 modules (M1–M8) compared against 1,000 length- and rate-matched null pseudo-panels sampled from 298 control genes (`h8_null_modules.tsv`). **Bottom panel:** Empirical aBSREL positive-selection rate on the *M. undulatus* terminal branch vs. Psittaculidae stem across the panel and control genes (`6_absrel_base_rate.tsv`).

Module	$n_{\textbf{genes}}$	**Prop. Relaxed**	**Null Mean**	p **(Prop.)**	**Mean** $\log_2 k$	**Null Mean**	p $(\log_2 k)$
M1	9	0.000	0.072	1.000	-0.100	-0.713	0.593
M2	6	0.333	0.057	0.092	-3.688	-0.489	0.024
M3	10	0.100	0.041	0.663	-1.374	-0.005	0.094
M4	11	0.000	0.088	0.805	-0.046	-0.792	0.378
M5	8	0.125	0.065	0.803	0.348	-0.679	0.188
M6	13	0.000	0.035	1.000	-0.491	-0.298	0.713
M7	8	0.250	0.084	0.328	-1.135	-0.742	0.641
M8	22	0.000	0.027	1.000	-0.272	0.038	0.585

Gene Set	**Tested**	**Melo** $p < 0.05$	**Melo** $q < 0.10$	**Stem** $p < 0.05$	**Stem** $q < 0.10$	**Fisher** p **vs. Ctrl** $(p < 0.05)$	**Fisher** p $(q < 0.10)$
H8 Full Panel (burgio2019, 87 genes)	86	6/86 (7.0%)	6/86 (7.0%)	1/86 (1.2%)	0/86 (0.0%)	1.0000	0.5888
H8 Module M7 Telomere/Shelterin (8 genes)	8	5/8 (62.5%)	5/8 (62.5%)	1/8 (12.5%)	0/8 (0.0%)	0.0037	6.94×10^{-4}
H8 Non-M7 Modules M1-M6, M8 (79 genes)	78	1/78 (1.3%)	1/78 (1.3%)	0/78 (0.0%)	0/78 (0.0%)	–	–
Genome-wide Control Pool (19 genes tested with aBSREL)	19	1/19 (5.3%)	0/19 (0.0%)	0/19 (0.0%)	0/19 (0.0%)	1.0000	1.0000

Table S4. Red-Team Sensitivity Audit of Phase 3 Genome-Wide Coupling ($r = 0.666$)

Summary of all stress tests applied to the 7,756-gene `RERconverge` correlation coupling between relative longevity (Rho_{LE}) and relative brain mass ($\text{Rho}_{\text{Brain}}$): seed reproducibility, Brownian motion null permulations, technical confounders, zero-branch invariance stratification, leave-one-clade-out perturbations, and true partial orthogonalization.

Red-Team Audit Condition / Perturbation	$r_{\textbf{tip}}$	$r_{\textbf{branch path}}$	$r(\textbf{Rho}_1, \textbf{Rho}_2)$	**Key Note**
A. Negative Controls & Brownian Permulation Nulls (`2_coupling_null_and_controls.tsv`)				
Observed: rel_log_life_expectancy vs rel_log_brain (saved	0.4783	0.4154	0.6663	r_Rho = 0.6663; branch_path_cor = 0.4
Observed (50sp matched RER): rel_log_life_expectancy vs re	0.4783	0.4253	0.7478	r_Rho = 0.7478; branch_path_cor = 0.4
Control 1: rel_log_life_expectancy vs log_body_mass (n=58)	0.4102	0.2885	0.6792	r_Rho = 0.6792; branch_path_cor = 0.2
Control 2: rel_log_life_expectancy vs clutch_size raw (n=5	-0.4345	-0.3482	-0.6280	r_Rho = -0.6280; branch_path_cor = -0.
Control 3: rel_log_life_expectancy vs rel_clutch_size (res	-0.1725	-0.2145	-0.1024	r_Rho = -0.1024; branch_path_cor = -0.
Control 4: rel_log_life_expectancy vs rel_incubation_days	0.2060	0.0513	0.3417	r_Rho = 0.3417; branch_path_cor = 0.0
Null 2A: Independent permulated traits (200 pairs)	0.0000	-0.0021	-0.0090	mean=-0.0090, SD=0.2885, mean|r|=0.2391, 95
Null 2B: Jointly simulated coupled BM traits (200 pairs, e	-0.2786	-0.2762	-0.2603	mean=-0.2603, SD=0.3388, 95%=[-0.7910, 0.453
Null 2C: Observed LE fixed vs 200 permulated Brain traits	0.0000	0.0326	0.0187	mean=0.0187, SD=0.3504, mean|r|=0.3002, 95%
B. Technical Confounder Adjustment & Zero-Branch Strata (`3_coupling_after_confounder_removal.tsv`)				
Raw unadjusted Rho (all 7,756 genes) ($n = 7756$)	–	–	0.6663	Spearman=0.6323; $r(\lvert\text{Rho}\rvert)$=0.54
OLS residuals after removing 7 linear+log technical covariates ($n = 7756$)	–	–	0.6539	Spearman=0.6219; $r(\lvert\text{Rho}\rvert)$=0.51
OLS residuals after removing 14 quadratic+log technical covariates ($n = 7756$)	–	–	0.6532	Spearman=0.6222; $r(\lvert\text{Rho}\rvert)$=0.53
Stratified by aln_codons: Q1 (lowest) ($n = 1940$)	–	–	0.6856	Spearman=0.6430; $r(\lvert\text{Rho}\rvert)$=0.60
Ultra-conserved stratum ($>$ 80% zero branches, $n = 1,450$)	–	–	0.6761	**117/131 (89.3%)** of shared hit
Informative stratum ($\geq$ 20 nonzero branches, $n = 6,182$)	–	–	0.6626	Only **14/131** shared hits
Most informative ($\geq$ 70% nonzero branches, $n = 1,263$)	–	–	0.6982	Only **2** shared hits
C. Leave-One-Clade-Out Sensitivity (`4_leave_one_clade_out.tsv`)				
All_50_species_with_brain ($n = 51$)	0.4783	0.4253	0.7478	PTEN=-0.395; MAD2L2=0.420
Drop_Strigopidae (Strigops + Nestor, -2 sp) ($n = 50$)	0.5095	0.4197	0.7646	PTEN=-0.379; MAD2L2=0.490
Drop_Cacatuidae (Cacatua, Calyptorhynchus, Eolophus, Probo ($n = 44$)	0.4631	0.4492	0.7626	PTEN=-0.396; MAD2L2=0.425
Drop_Loriini (Chalcopsitta, Eos, Trichoglossus, -5 sp) ($n = 49$)	0.4773	0.4245	0.7483	PTEN=-0.385; MAD2L2=0.419
Drop_Melopsittacus_undulatus (-1 sp) ($n = 50$)	0.4458	0.4048	0.7285	PTEN=-0.378; MAD2L2=0.401
Drop_All_Psittaculidae (-13 sp, leave 37 non-Psittaculidae ($n = 38$)	0.1955	0.2095	0.5225	PTEN=-0.356; MAD2L2=0.344
Drop_Arini_Macaws_Conures (Ara, Anodorhynchus, Pyrrhura, e ($n = 33$)	0.3798	0.2407	0.6489	PTEN=-0.355; MAD2L2=0.335
Drop_Androglossini (Amazona, Pionus, Pyrilia, Triclaria, - ($n = 48$)	0.5419	0.5053	0.7975	PTEN=-0.421; MAD2L2=0.425

Red-Team Audit Condition / Perturbation	$r_{\mathbf{tip}}$	$r_{\mathbf{branch\ path}}$	$r(\mathbf{Rho}_1, \mathbf{Rho}_2)$	**Key Note**
D. Orthogonalization of Brain Beyond Longevity (`5_residual_brain_summary.tsv`)				
1. Unadjusted rel_log_brain (50 sp)	0.4783	0.4253	0.6663	50sp=0.7478; cor(Br)=1.000
2. PGLS residual rel_brain_given_le (scripts/35_prepare_re	0.4269	0.3704	0.6385	50sp=0.7163; cor(Br)=0.999
3. Tip-OLS orthogonalized rel_brain_given_le_OLS (beta_OLS	-0.0000	-0.0631	0.3322	50sp=0.3691; cor(Br)=0.875
4. Branch-path orthogonalized paths_brain_orth_le (branch_	–	-0.0000	0.3888	50sp=0.4332; cor(Br)=0.909
5. Per-gene analytical partial correlation r(RER_g, P_Brai	–	0.0000	0.4135	50sp=0.4612; cor(Br)=0.934

Table S5. Dual-Tier Phase 3 Candidate Gene Catalog: Ultra-Conserved (Tier A) vs. Informative (Tier B)

Tier A lists the 13 flagship shared candidate genes from the original Phase 3 synthesis together with their fraction of zero-length branches ($\leq 10^{-6}$), branch-orthogonalized $\text{Rho}_{\text{Brain}\perp\text{LE}}$, OLS tip-orthogonalized permulation p, and 30-split cross-validation sign concordance. **Tier B** lists all 14 shared genes passing $P < 0.05$ and permP < 0.05 for both traits that have ≥ 20 nonzero branches.

Gene	% Zero Branches (n_{nz})	Rho_{LE}	permP_{LE}	$\text{Rho}_{\text{Brain}}$	$\text{Rho}_{\text{Br}\perp\text{LE}}$ (Orth.)	OLS permP_{Br}	CV Sign Conc.
PTEN	99.1% (1)	-0.374	0.0080	-0.424	-0.342	0.109	100% / 9
CTNNB1	99.1% (1)	-0.439	0.0040	-0.459	-0.355	0.228	100% / 1
CTNNBIP1	99.0% (1)	-0.361	0.0200	-0.443	-0.374	0.149	97% / 9
PSMA4	99.1% (1)	-0.426	0.0040	-0.464	-0.361	0.188	100% / 9
SSU72	98.9% (1)	-0.465	0.0060	-0.470	-0.374	0.168	100% / 1
TOMM20	96.9% (2)	-0.435	0.0359	-0.498	-0.377	0.267	97% / 10
ANAPC11	96.2% (3)	-0.406	0.0060	-0.434	-0.370	0.040	97% / 9
MAD2L2	99.0% (1)	+0.418	0.0120	+0.534	+0.477	0.030	100% / 9
HSPA8	98.2% (2)	+0.404	0.0080	+0.354	+0.277	0.198	100% / 9
RTN1	97.3% (3)	+0.516	0.0080	+0.440	+0.309	0.188	100% / 9
SOX14	98.1% (2)	+0.476	0.0040	+0.500	+0.384	0.119	100% / 1
NCALD	98.2% (2)	+0.417	0.0080	+0.493	+0.401	0.099	100% / 9
CBLN4	96.7% (3)	+0.456	0.0040	+0.401	+0.275	0.366	100% / 9
TKT	60.9% (27)	+0.408	0.0040	+0.347	–	–	–
PHOSPHO1	52.3% (53)	-0.261	0.0120	-0.267	–	–	–
TADA2A	72.6% (20)	+0.316	0.0220	+0.334	–	–	–
LOC101877632	50.8% (32)	-0.331	0.0040	-0.321	–	–	–
LOC101872892	44.1% (62)	+0.239	0.0319	+0.384	–	–	–
SLC4A7	44.8% (37)	+0.296	0.0319	+0.304	–	–	–
DEPDC4	21.5% (73)	-0.250	0.0100	-0.214	–	–	–
HUS1	54.1% (51)	-0.222	0.0120	-0.253	–	–	–
LOC101868077	66.3% (34)	+0.231	0.0479	+0.276	–	–	–
MCEE	60.2% (41)	+0.219	0.0359	+0.279	–	–	–
C11H17orf75	45.2% (51)	-0.226	0.0479	-0.279	–	–	–
MAP3K8	79.8% (22)	+0.200	0.0439	+0.254	–	–	–
DIPK1C	41.4% (58)	+0.204	0.0240	+0.252	–	–	–
LOC101870278	21.1% (45)	-0.263	0.0240	-0.274	–	–	–

Table S6. Exhaustive Statistical Inventory (78,802 Tests), Null Findings, and AlphaFold 3D Summary

Phase / Scope	Tests Run	Correction	Positive / Nominally Significant	Null / Negative Findings Reported
Phase 0 & 0b (Allometry, Captive/Wild, Brain & Life-History PGLS)	48	Model AICc + PGLS t-tests (Supertree + Smith2024)	Body mass -> longevity (p=4.2e-18); Relative brain -> longevity (p=0.0012); Captive vs wild longevity r=0.91	Diet guild effect on longevity after mass+phylogeny (NULL, p>0.25); Latitude/insularity effect on residual longevity (NULL); OU single-optimum vs multi-regime outside Psittaculidae/Cacatuidae/Strigopidae (NULL)
Phase 1 (Assembly Quality, Mito Identity, Mash Distance, Cancer Necropsy, Measurement Error)	34	PGLS + spearman/pearson + ME sensitivity	Assembly quality (N50, BUSCO) is uncorrelated with longevity (validating no quality bias); M. undulatus high cancer prevalence in veterinary necropsy series	Assembly scaffold N50 vs residual longevity (NULL, r=0.04, p=0.76); Mitochondrial substitution rate vs longevity (NULL, p>0.30); Genome-wide Mash k-mer divergence vs longevity (NULL after tree height correction)
Phase 2 Pilot & Pilot-Plus (12-species and 17-species exploratory runs)	348	BH-FDR across 87 panel genes (12sp and 17sp pilots)	Early signal of higher k in M4 telomere in 17-species pilot	Module-level proportion of relaxed genes (k<1, q<0.10) across modules in 12sp and 17sp pilots (NULL); No pseudogenization/loss of TP53, RB1, PTEN, or POT1 across long-lived parrots (NULL)
Phase 2 Main (58 species, 101-gene panel: Integrity, RELAX, aBSREL, 298-gene Null Pool)	812	BH-FDR across 87 panel genes per tree (burgio2019 & smith2024) + 10,000 module label permutations + 1,000 stratified null pseudo-panels from 298 control genes	Module heterogeneity in prop_relaxed on burgio2019 (p=0.0229, NOT replicated on smith2024 p=0.1271); M7 Telomere 5/8 genes with aBSREL q<0.10 on M. undulatus branch in original MAFFT (collapses on window cleaning / multi-hit); TERT intensified k (q=0.0008/0.011, 15 sp)	H8a Gene loss / pseudogenization in Psittaculidae vs rest (NULL: 100% of candidate disruptions were assembly fragmentation/coverage artifacts or heterozygous); H8b Module heterogeneity in mean log2(k) (NULL: p=0.1643 burgio2019, p=0.9551 smith2024); M2 mean log2(k) p=0.024 vs null (NULL after 16-test correction and median sensitivity p=0.07); Modules M1, M3-M8 vs 298-gene null (ALL NULL: p=0.09 to 1.00); H8c aBSREL positive selection on the Psittaculidae stem branch (NULL: 0/87 panel genes had q<0.10 on stem)
Phase 3 Genome-Wide RERconverge (7,756 genes x 2 trees x 3 H6 traits + 2 H8 within-clade runs + GO enrichment)	77,560	BH-FDR on parametric P and BH-FDR on 1,000 Brownian motion permulation P-values (permQ) across 7,756 genes; Wilcoxon GMT enrichment across GO_BP terms	Top individual gene rankings (PTEN P=9.76e-5, permP=0.001; MAD2L2, HSPA8); 131-151 genes with nominal P<0.05 & permP<0.05 in both longevity and brain	Genome-wide BH-FDR across all 7,756 genes for single genes (NULL at permQ < 0.10: minimum permP is bounded by 1/1001 = 0.000999 with 1,000 permulations, so single-gene permQ cannot reach <0.10 without >=100,000 permulations); H8 within-Psittaculidae RERconverge on 86 panel genes after BH correction (NULL: 0/86 genes had panelQ < 0.10); H8 within-rest RERconverge on 86 panel genes after BH correction (NULL: 0/86 genes had panelQ < 0.10; PTEN was nominal P=1.18e-4, panel_padj_analytic=0.010, but permulation panelQ=0.145)

Protein	UniProt ID	AA Len.	Global pLDDT	Spp.	Structural Mapping Finding (`structural_3d_summary.tsv`)
`POT1`	A0A8C6K5H8 (M. undulatus) / Q9NUX5 (Human)	780	79.21	8	58 M. undulatus-specific substitutions distributed across ssDNA-binding OB1/OB2 (34 sites) and TPP1-interacting OB3/HJRL C-terminus (16 sites); mean pLDDT at mutated sites = 72.6
`RTEL1`	Q9NZ71 (Canonical) / A0A8V5GYQ7 (M. undulatus)	1219	71.67	23	158 M. undulatus-specific substitutions (aBSREL q=1.5e-8): 43 in the Rad3/FeS/Arch helicase core and 115 in the C-terminal Harmonin-N / PCNA-interacting region
`PTEN`	A0A8C6JLV9 (M. undulatus) / P60484 (Human)	403	82.47	57	100% invariant catalytic P-loop (H123-C124-K125-A126-G127-K128-G129-R130, mean pLDDT=93.0) and near-zero divergence across the rigid Phosphatase+C2 superdomain (residues 14-350); polymorphisms restricted to the disordered C-terminal tail (residues 351-403, pLDDT=35.1)
`MAD2L2`	Q9UI95 (REV7)	211	90.39	54	High-confidence globular HORMA fold (global pLDDT=90.4); evolutionary rate acceleration (Rho_brain=+0.53/+0.57, #1 genome-wide) maps to surface-exposed and safety-belt hinge residues while preserving the hydrophobic core